

MATH 2R03 - Assignment 1

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Question 1

Show that:

(a) $(\alpha + \beta) + \lambda = \alpha + (\beta + \lambda)$ for all $\alpha, \beta, \lambda \in \mathbb{C}$.

Solution:

Let $\alpha = \alpha_1 + \alpha_2 i, \beta = \beta_1 + \beta_2 i, \lambda = \lambda_1 + \lambda_2 i$, where $\alpha_1, \alpha_2, \beta_1, \beta_2, \lambda_1, \lambda_2 \in \mathbb{R}$. We therefore have that:

$$\begin{aligned}
 (\alpha + \beta) + \lambda &= (\alpha_1 + \alpha_2 i + \beta_1 + \beta_2 i) + \lambda_1 + \lambda_2 i && \text{(By definition of } \alpha, \beta, \lambda) \\
 &= (\alpha_1 + \beta_1 + \alpha_2 i + \beta_2 i) + \lambda_1 + \lambda_2 i && \text{(By commutativity of addition)} \\
 &= (\alpha_1 + \beta_1 + (\alpha_2 + \beta_2)i) + \lambda_1 + \lambda_2 i && \text{(By distributivity of multiplication over addition)} \\
 &= (\alpha_1 + \beta_1 + (\alpha_2 + \beta_2)i) && \\
 &= \alpha_1 + \beta_1 + (\alpha_2 + \beta_2)i + \lambda_1 + \lambda_2 i && \text{(By definition of addition over } \mathbb{C}) \\
 &= ((\alpha_1 + \beta_1) + \lambda_1) + ((\alpha_2 + \beta_2) + \lambda_2)i && \text{(By definition of addition over } \mathbb{C}) \\
 &= (\alpha_1 + (\beta_1 + \lambda_1)) + (\alpha_2 + (\beta_2 + \lambda_2))i && \text{(By associativity of addition)} \\
 &= \alpha_1 + (\beta_1 + \lambda_1) + \alpha_2 i + (\beta_2 + \lambda_2)i && \text{(By distributivity)} \\
 &= \alpha_1 + \alpha_2 i + (\beta_1 + \lambda_1) + (\beta_2 + \lambda_2)i && \text{(By commutativity of addition)} \\
 &= \alpha + (\beta + \lambda) && \text{By definition of } \alpha, \beta, \lambda
 \end{aligned}$$

(b) $(\alpha + \beta) + \lambda = \alpha + (\beta + \lambda)$ for all $\alpha, \beta, \lambda \in \mathbb{Z}/2\mathbb{Z}$.

Solution:

In the case that $\alpha, \beta, \lambda \in \mathbb{Z}/2\mathbb{Z}$, we have that:

$$(\alpha + \beta) + \lambda = \alpha + (\beta + \lambda)$$

Since $\mathbb{Z}/2\mathbb{Z}$ uses regular integer addition with the exception that $1 + 1 := 0$. Since this is still associative, $(\alpha + \beta) + \lambda = \alpha + (\beta + \lambda)$ for all $\alpha, \beta, \lambda \in \mathbb{Z}/2\mathbb{Z}$.

Question 2

Show that:

(a) $\lambda(\alpha + \beta) = \lambda\alpha + \lambda\beta$ for all $\lambda, \alpha, \beta \in \mathbb{C}$.

Solution:

As earlier, letting $\alpha = \alpha_1 + \alpha_2 i$, $\beta = \beta_1 + \beta_2 i$, and $\lambda = \lambda_1 + \lambda_2 i$ respectively, we have that:

$$\lambda(\alpha + \beta) = (\lambda_1 + \lambda_2 i)((\alpha_1 + \alpha_2 i) + (\beta_1 + \beta_2 i))$$

(By definition of α, β, λ)

$$= \lambda_1((\alpha_1 + \alpha_2 i) + (\beta_1 + \beta_2 i)) + \lambda_2 i((\alpha_1 + \alpha_2 i) + (\beta_1 + \beta_2 i))$$

(By distributivity of multiplication over addition)

$$= \lambda_1(\alpha_1 + \alpha_2 i) + \lambda_1(\beta_1 + \beta_2 i) + \lambda_2 i(\alpha_1 + \alpha_2 i) + \lambda_2 i(\beta_1 + \beta_2 i)$$

(By distributivity of multiplication over addition)

$$= \lambda_1(\alpha_1 + \alpha_2 i) + \lambda_2 i(\alpha_1 + \alpha_2 i) + \lambda_1(\beta_1 + \beta_2 i) + \lambda_2 i(\beta_1 + \beta_2 i)$$

(By commutativity of addition)

$$= (\lambda_1 + \lambda_2 i)(\alpha_1 + \alpha_2 i) + (\lambda_1 + \lambda_2 i)(\beta_1 + \beta_2 i)$$

(By distributivity of multiplication over addition)

$$= \lambda\alpha + \lambda\beta$$

(By definition of α, β, λ)

(b) $\lambda(\alpha + \beta) = \lambda\alpha + \lambda\beta$ for all $\lambda, \alpha, \beta \in \mathbb{Z}/2\mathbb{Z}$.

Solution:

Since multiplication is defined as usual, on $\mathbb{Z}/2\mathbb{Z}$, and integer multiplication is distributive, we have that

$$\lambda(\alpha + \beta) = \lambda\alpha + \lambda\beta$$

for all $\lambda, \alpha, \beta \in \mathbb{Z}/2\mathbb{Z}$.

Question 3

Show that:

(a) For every $\alpha \in \mathbb{C}$ with $\alpha \neq 0$, there exists a unique $\beta \in \mathbb{C}$ such that $\alpha\beta = 1$.

Solution:

Let $\alpha = a + bi$, where $\alpha \neq 0$. I assert that $\beta = \frac{1}{a+bi}$ would satisfy $\alpha\beta = 1$, to prove $\beta \in \mathbb{C}$, consider that, since $\alpha \neq 0$, $a \neq 0 \vee b \neq 0$ and thus $\frac{1}{a+bi}$ is defined. Therefore, we can see that:

$$\frac{1}{a+bi} = \frac{1}{a+bi} \cdot 1 \quad (\text{By multiplicative identity})$$

$$= \frac{1}{a+bi} \cdot a - bia - bi \quad (\text{By multiplicative inverse})$$

$$= \frac{a - bi}{a^2 - (bi)^2} \quad (\text{By usual rules of multiplication})$$

$$= \frac{a - bi}{a^2 + b^2} \quad (\text{By definition of } i. \text{ Note that } \alpha \neq 0 \implies a \neq 0 \vee b \neq 0 \implies a^2 + b^2 > 0)$$

$$= \frac{a}{a^2 + b^2} + \frac{-b}{a^2 + b^2}i \quad (\text{By usual rules of addition})$$

Thus, since $\frac{a}{a^2+b^2}, \frac{-b}{a^2+b^2} \in \mathbb{R}$, $\frac{1}{a+bi} \in \mathbb{C}$ and therefore $\beta = \frac{1}{a+bi}$ satisfies $b \in \mathbb{C}$ such that $\alpha\beta = 1$ when $\alpha \neq 0$.

To prove that the multiplicative inverse of α is unique, simply consider the case that $\alpha\beta_1 = 1$ and $\alpha\beta_2 = 1$ where $\beta_1, \beta_2 \in \mathbb{C}$. Then, we have that:

$$\begin{aligned}\beta_1 &= 1 \cdot \beta_1 && \text{(By multiplicative identity)} \\ &= (\alpha\beta_2)\beta_1 && \text{(By definition of } \beta_2) \\ &= (\beta_2\alpha)\beta_1 && \text{(By commutativity of multiplication)} \\ &= \beta_2(\alpha\beta_1) && \text{(By associativity of multiplication)} \\ &= \beta_2 \cdot 1 && \text{(By definition of } \beta_1) \\ &= \beta_2 && \text{(By multiplicative identity)}\end{aligned}$$

Thus, $\beta_1 = \beta_2$ and therefore the multiplicative inverse of α must be unique.

(b) For every $\alpha \in \mathbb{Z}/2\mathbb{Z}$ with $\alpha \neq 0$, there exists a unique $\beta \in \mathbb{Z}/2\mathbb{Z}$ such that $\alpha\beta = 1$.

Solution:

If $\alpha \in \mathbb{Z}/2\mathbb{Z}$, then $\alpha = 0 \vee \alpha = 1$. Thus, if $\alpha \neq 0$, then $\alpha = 1$. Thus, selecting $\beta = 1$, we have that $\beta \in \mathbb{Z}/2\mathbb{Z}$ and:

$$\begin{aligned}\alpha\beta &= 1 \cdot 1 \\ &= 1\end{aligned}$$

Note that only $\beta = 1$ can possibly be the multiplicative inverse of such an α since, if $\beta = 0$:

$$\begin{aligned}\alpha\beta &= 1 \cdot 0 \\ &= 0\end{aligned}$$

Thus, the multiplicative inverse in of any non-zero α in $\mathbb{Z}/2\mathbb{Z}$ must be unique.

Question 4

Show that $(ab)x = a(bx)$ for all $x \in \mathbb{F}^n$, where $\mathbb{F}$ is any field.

Solution:

Let $x = (x_1, x_2, \dots, x_n)$. We therefore have that:

$$\begin{aligned}(ab)x &= (ab)(x_1, x_2, \dots, x_n) && \text{(By definition of } x\text{)} \\ &= ((ab)x_1, (ab)x_2, \dots, (ab)x_n) && \text{(By definition of scalar multiplication over } n\text{-tuples)} \\ &= (a(bx_1), a(bx_2), \dots, a(bx_n)) && \text{(By associativity of multiplication)} \\ &= a(bx_1, bx_2, \dots, bx_n) && \text{(By definition of scalar multiplication over } n\text{-tuples)} \\ &= a(bx) && \text{(By definition of } x \text{ and scalar multiplication over } n\text{-tuples)}\end{aligned}$$

Question 5

Show that $(\alpha + \beta)x = \alpha x + \beta x$ for all $\alpha, \beta \in \mathbb{F}$ and all $x \in \mathbb{F}^n$, where $\mathbb{F}$ is any field.

Solution:

Let $x = (x_1, x_2, \dots, x_n)$. We therefore have that:

$$\begin{aligned}(\alpha + \beta)x &= (\alpha + \beta)(x_1, x_2, \dots, x_n) && \text{(By definition of } x\text{)} \\&= ((\alpha + \beta)x_1, (\alpha + \beta)x_2, \dots, (\alpha + \beta)x_n) && \text{(By definition of scalar multiplication over } n\text{-tuples)} \\&= (\alpha x_1 + \beta x_1, \alpha x_2 + \beta x_2, \dots, \alpha x_n + \beta x_n) && \text{(By distributive property)} \\&= (\alpha x_1, \alpha x_2, \dots, \alpha x_n) + (\beta x_1, \beta x_2, \dots, \beta x_n) && \text{(By definition of addition over } n\text{-tuples)} \\&= \alpha(x_1, x_2, \dots, x_n) + \beta(x_1, x_2, \dots, x_n) && \text{(By definition of scalar multiplication over } n\text{-tuples)} \\&= \alpha x + \beta x && \text{(By definition of } x\text{)}\end{aligned}$$